

Project Requirement Documentation

Project Name: Navigators

SIH Problem Statement ID: SIH26168 (S. No. 168)

Organization: Indian Space Research Organisation (ISRO)

1. Introduction

The objective of this project is to develop an AI/ML-based Intelligent Dead Reckoning (IDR) system for seamless navigation. GNSS/GPS-based navigation becomes unreliable in tunnels, underpasses, and urban environments. When GNSS is lost, conventional navigation often fails. This solution maintains continuous and accurate vehicle positioning using inertial sensing and intelligent estimation.

2. Core Objective

Develop an AI/ML-powered IDR system that utilizes a smartphone's IMU sensors to continue vehicle navigation during GNSS outages without relying on OBD-II or a vehicle speedometer. The solution combines AI/ML, IMU processing, GNSS/INS fusion, road constraints, and map matching in a lightweight edge/mobile deployment.

3. Key Technical Requirements

- **AI-based speed estimation:** Estimate vehicle speed/velocity from smartphone IMU data without OBD-II.
- **Vibration and noise filtering:** Handle engine vibration, road vibration, potholes, bumps, shocks, and phone movement.
- **Phone alignment:** Automatically estimate phone pitch, roll, and yaw relative to vehicle motion.
- **Map matching:** Constrain drifting inertial trajectories to plausible roads using offline maps and road-network information.
- **Vehicle kinematics:** Use Non-Holonomic Constraints (NHC) and related motion constraints to reduce impossible trajectories.
- **GNSS + INS fusion:** Fuse GNSS and inertial measurements while reducing drift and improving velocity/position estimates.
- **Seamless outage handling:** Switch rapidly between GNSS-aided navigation and dead reckoning, then return to fused navigation when GNSS returns.
- **External IMU support:** Core edge engine should accommodate external IMU data, not only smartphone sensors.
- **100% Offline Edge Execution:** The entire AI model and navigation engine must run directly on the mobile device's CPU/NPU without requiring *any* backend servers, internet connection, or cellular network.
- **WebAssembly/ONNX Integration:** Utilize ONNX Runtime Web to execute PyTorch neural networks locally within a Progressive Web App (PWA).
- **Real-time navigation UI:** Display a continuous vehicle position without obvious freezing or location jumps.

4. Performance Targets

- **Dead-reckoning positional drift:** < 10% of the distance travelled

- **50 m GNSS-denied travel:** < 5 m drift, in under 1 minute
- **1 km GNSS-denied travel:** < 100 m drift at 60 km/h
- **Smartphone GNSS + INS update rate:** 10 Hz
- **FOG-IMU edge engine capability:** Around 200 Hz or higher

Navigators IDR: Comprehensive Project Guide

Project: Navigators (SIH26168 for ISRO)

Focus: AI/ML-based Navigation System during GNSS Denial

1. The Problem We Are Solving

In mission-critical scenarios—such as military operations, disaster response, and aerospace navigation (ISRO)—vehicles rely heavily on Global Navigation Satellite Systems (GNSS) like GPS or NavIC. However, GNSS signals are easily compromised. They can be jammed by adversaries, blocked by natural terrain (tunnels, dense forests), or fail due to atmospheric conditions.

When a vehicle enters a "GNSS-denied" environment, traditional navigation systems fail, leaving operators blind. While professional Inertial Navigation Systems (INS) exist, they are extremely expensive and bulky. Modern vehicles and personnel carry smartphones with built-in sensors (accelerometers and gyroscopes), but these commercial sensors are extremely noisy. If you simply integrate mobile sensor data to find position (traditional Dead Reckoning), the error multiplies exponentially, leading to severe drift within seconds.

2. Our Solution: Intelligent Dead Reckoning (IDR) Edge AI

We have built a 100% offline, AI-powered navigation engine that runs directly on commercial mobile devices (Edge Inference) without requiring any internet connection.

How it works:

1. **AI Velocity Estimation:** Instead of relying on pure physics equations that suffer from sensor noise, we trained a Temporal Convolutional Network (TCN). This neural network analyzes a rolling window of vibration and motion patterns from the phone's sensors to accurately predict the vehicle's speed and heading.
2. **Physical Constraints:** The AI is fused with physical constraints like Zero Velocity Updates (ZUPT) and Non-Holonomic Constraints (NHC) to mathematically eliminate drift when the vehicle is stopped or moving in a straight line.
3. **Seamless Handover:** When GPS is lost, the system instantly switches to AI Dead Reckoning. When GPS returns, it automatically corrects any minor drift and resumes satellite tracking.
4. **100% Offline Edge Architecture:** The entire neural network (compiled via ONNX) and Dead Reckoning engine run in WebAssembly directly on the mobile phone's CPU. **No backend servers or cloud connections are required.**

3. Impact

- **Resilience:** Provides uninterrupted navigation in tunnels, urban canyons, and jammed environments.
- **Cost-Efficiency:** Achieves professional-grade navigation tracking using cheap, ubiquitous commercial smartphone sensors.
- **Security & Privacy:** Because the system operates entirely offline on the edge device, zero location

data is transmitted over a network, making it highly secure for defense and space agency applications.

4. How We Will Scale It

- **Model Generalization:** The current AI model can be retrained with massive datasets (like IO-VNBD) covering different vehicle types (cars, trucks, rovers) and mounting positions. As the model sees more data, its accuracy scales universally.
- **Hardware Acceleration:** By utilizing WebAssembly and ONNX Runtime, the system automatically scales to take advantage of whatever hardware it is running on (e.g., utilizing a phone's Neural Processing Unit (NPU) or GPU for faster, lower-power inference).
- **Platform Agnostic:** Because the core engine is built using Web Standards and ONNX, the exact same AI model and logic can run on an iPhone, an Android tablet, a Raspberry Pi inside a drone, or a dashboard infotainment system without rewriting the core engine.

5. How We Will Distribute It

We have architected the solution for frictionless, decentralized distribution:

1. **Progressive Web App (PWA):** The application is hosted as a secure webpage. Users simply navigate to the URL, and the app installs itself (along with the AI model) directly onto the device's home screen. This bypasses the need for Apple App Store or Google Play Store approvals, which is crucial for internal government/ISRO distribution.
2. **Automatic Background Updates:** If a new, more accurate AI model (model.onnx) is trained by the central team, the PWA's Service Worker will silently download it the next time the device touches Wi-Fi. The operator doesn't have to manually "update" the app.
3. **Native Wrappers:** For highly secure, air-gapped deployments, the entire HTML/JS/ONNX bundle can be wrapped into a native Android .apk using tools like Capacitor, allowing it to be sideloaded via USB onto military or ISRO field devices.